

Intubation with Missing BVM



Section I: Scenario Demographics

Scenario Title:	Intubation with Missing BVM		
Date of Development:	13/12/2017 (update)		
Target Learning Group:	☒ Juniors (PGY 1 – 2)	☐ Seniors (PGY ≥ 3)	☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Dr. N Kester-Greene / Dr A Petrosoniak
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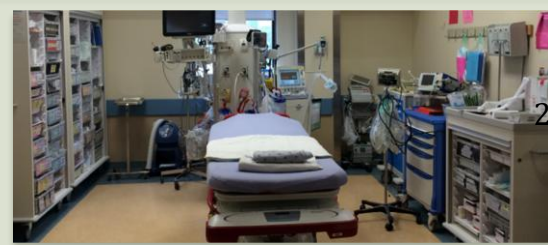
Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To allow learners to practice an approach to RSI in a scenario where there is a problem with the equipment
CRM Objectives:	1. Communicate effectively with team regarding a plan for RSI 2. Ensure all preparation is complete before initiating a stressful, high-risk procedure
Medical Objectives:	1. Initiate early sepsis management including ABC support, fluids, and antibiotics 2. Recognize the need for rapid sequence intubation in a hypoxic patient 3. Identify equipment, medication and personnel needed for RSI 4. Perform RSI in a septic, hypoxic patient

Case Summary: Brief Summary of Case Progression and Major Events
A 41-year old male with HIV (not on HAART) presents to the ED with a cough for 10 days, progressive dyspnea and fever. He is hypoxic at triage and brought immediately to the resuscitation room. He has transient improvement on oxygen but then has progressive worsening of his hypoxia and dyspnea. Intubation is required. The team needs to prepare for RSI and identify that the BVM is missing from the room prior to intubation. Depending on the experience of the participants, the sim instructor may choose to give the team leader a local “pre-intubation checklist” prior to starting the scenario.

References
Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). <i>Rosen's emergency medicine: Concepts and clinical practice</i> . St. Louis: Mosby.
Nickson, C. (2015). Rapid sequence Intubation (RSI). <i>Life in the Fast Lane</i> . Retrieved from: https://lifeinthefastlane.com/ccr/rapid-sequence-intubation/
PulmCCM. (2013). Intubation Checklists in the ICU and ED. Can They Save Lives? <i>PulmCCM</i> . Retrieved from: https://pulmccm.org/review-articles/emcrits-intubation-checklist-is-it-bigger-than-yours/
Hardy, G., & Horner, D. (2016). Best Bets: Should Real Resuscitationists Use Airway Checklists? <i>Emerg Med J</i> . Vol 33(6):439-41.

Intubation with Missing BVM



Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☒ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
Confederates	Brief Description of Role		
Nurse	To assist at bedside and cue learners to patient's status.		
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☒ Defibrillator Pads	☒ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☒ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ *Bag Valve Mask* (removed from room prior to scenario)	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other:	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
One IV line in situ.			
E. Approximate Timing			
Set-Up:	3 min	Scenario:	12 min
		Debriefing:	15 min

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Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

You are working in a large community ED. The triage nurse tells you that she has just put a patient in the resuscitation room. He is a 41-year old man with HIV. He is known to be non-compliant with his anti-retrovirals. He noticed progressive shortness of breath over 3-4 days and has had a dry cough for 10 days. His O₂ sat was in the 80s at triage.

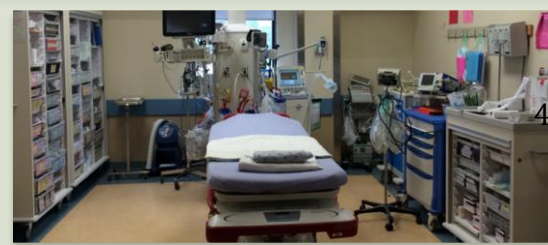
B. Patient Profile and History

Patient Name: Tony Schwartz		Age: 41		Weight: 80kg	
Gender: ☒ M ☐ F		Code Status: Full Code			
Chief Complaint: Dyspnea					
History of Presenting Illness: He noticed progressive shortness of breath over 3-4 days and has had a dry cough for 10 days. He thinks last night he had a fever. No recent travel. No sick contacts.					
Past Medical History:	HIV x 12 years		Medications:	None	
				No HAART (worried about side effects)	
Allergies: None					
Social History: Lives alone. Software tech. Occasional beer, MJ. No IVDU.					
Family History: Non-contributory					
Review of Systems:	CNS:	Nil			
	HEENT:	Nil			
	CVS:	Nil			
	RESP:	Tachypneic, increased work of breathing, cough			
	GI:	Nil			
	GU:	Nil			
	MSK:	Nil	INT:	Nil	

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display		☒ Monitor On, no data displayed		☐ Monitor on Standard Display	
HR: 105/min	BP: 110/70	RR: 30/min	O ₂ SAT: 84% on RA		
Rhythm: Sinus tach	T: 38.3°C	Glucose: 6.2mmol/L	GCS: 15 (E4 V5 M6)		
General Status: Alert but having respiratory distress. 2-3 word dyspnea.					
CNS:	Alert, able to give history.				
HEENT:	Nil				
CVS:	No murmur				
RESP:	Bilateral coarse crackles. Increased RR and WOB.				
ABDO:	Soft and non-tender				
GU:	Nil				
MSK:	Nil	SKIN:	Nil		

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Section VI: Scenario Progression

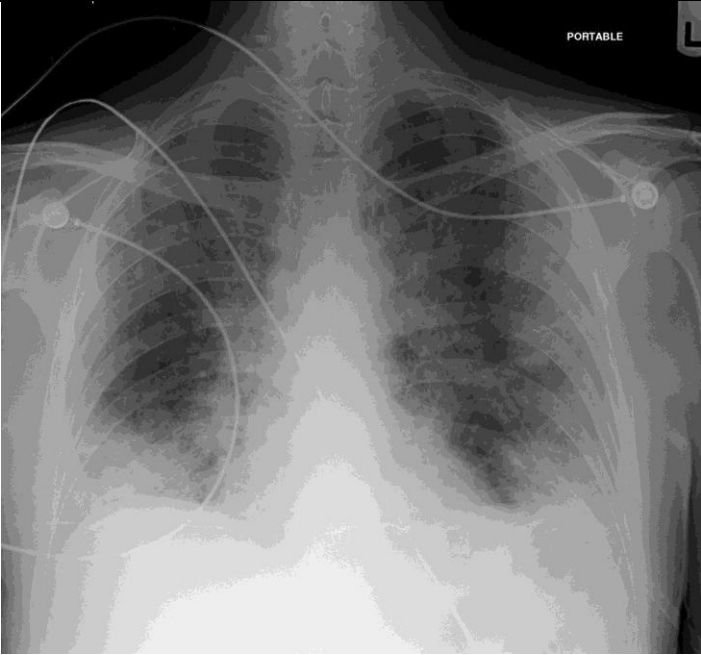
Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus HR: 105/min BP: 110/70 RR: 30/min O ₂ SAT: 84% T: 38.3°C	Able to answer questions. 2-3 word dyspnea.	<u>Learner Actions</u> ☐ Apply monitors ☐ Apply O ₂ (NRB) ☐ Obtain 2 nd IV ☐ Bloodwork including blood cultures, VBG, lactate ☐ IV saline 1-2L bolus ☐ Call for pCXR ☐ Give broad spectrum antibiotics +/- sepra	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - NRB placed → O ₂ 90% - NS bolus → BP 125/80 <u>Triggers</u> <i>For progression to next state</i> - All actions complete or 5 min into case → 2. Hypoxia
2. Hypoxia Rhythm: Sinus HR: 105/min BP: 110/70 RR: 30/min O ₂ SAT: 80%	SpO ₂ starts dropping despite mgmt. Patient progressively becomes more drowsy.	<u>Learner Actions</u> ☐ Call RT/BiPAP ☐ 2 nd IV (if not done) ☐ Airway assessment ☐ Ask for RSI meds ☐ Ask for RSI equipment ☐ Notice BVM is missing and request before proceeding ☐ +/- Nasal prongs for apneic oxygenation	<u>Modifiers</u> - O ₂ drops 2% per minute - If the hypoxia is not noticed, RN to prompt "his sats are getting really low" - If notice missing BVM → RN to retrieve one <u>Triggers</u> - RSI Meds given → 3. Intubation
3. Intubation Rhythm: Sinus HR: 105/min BP: 110/70 RR: 30/min O ₂ SAT: 65%	Acute drop in spO ₂ on induction	<u>Learner Actions</u> ☐ Bag patient with BVM ☐ Intubation ☐ Confirm tube placement	<u>Modifiers</u> - BVM → O ₂ 95% <u>Triggers</u> - Intubation attempt without using BVM first or no BVM at bedside → 4. PEA arrest - BVM then intubate → 5. Resolution
4. PEA arrest Rhythm: Junctional HR: 35 (no pulse) BP: 0 RR: BMV rate (10-12) O ₂ SAT: 0 EtCO ₂ : 10 (15-20 if high quality CPR)	Patient pulseless and non-responsive.	<u>Learner Actions</u> ☐ Recognize PEA arrest and start CPR ☐ Run resus per ACLS ☐ Epi 1mg q3-5min ☐ Intubation	<u>Modifiers</u> - If no intubation after epi x2, RN to prompt "maybe we should intubate the patient" <u>Triggers</u> - Intubation → 5. Resolution
5. Resolution Rhythm: sinus HR: 90/min BP: 110/75 RR: 12/min (vented) O ₂ SAT: 95% EtCO ₂ : 40	Patient intubated and sedated.	<u>Learner Actions</u> ☐ Call ICU ☐ Start sedation ☐ Call for pCXR	<u>Triggers</u> - Call ICU → END OF CASE

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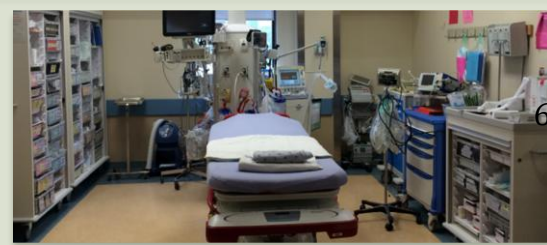
Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results	
No results provided during case.	

Images (ECGs, CXRs, etc.)	
	
(CXR source: https://radiopaedia.org/cases/35823)	

Ultrasound Video Files (if applicable)	
None	

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
Educational Goal:	To allow learners to practice RSI in a scenario where there is a problem with the equipment
CRM Objectives:	1. Communicate effectively with team before and during RSI 2. Ensure all preparation is complete before initiating a stressful, high-risk procedure
Medical Objectives:	1. Initiate early sepsis management including ABC support, fluids, and antibiotics 2. Recognize the need for rapid sequence intubation in a hypoxic patient 3. Identify equipment, medication and personnel needed for RSI 4. Perform RSI in a septic, hypoxic patient
Sample Questions for Debriefing	
1. What are the indications for intubation? Which were present for this patient? 2. What was your approach to intubation in this patient? What factors did you consider when choosing meds and equipment? 3. What are some ways to ensure that you are prepared for intubation? What tools/checklists have you seen or used? 4. How can the team members best work together during an intubation? 5. How does management of sepsis change in an immunocompromised patient? Are there any specific treatment differences to consider in HIV?	
Key Moments	
1. Recognize and treat sepsis in an immunocompromised patient	
2. Decision to intubate for hypoxic respiratory failure	
3. Assemble all equipment prior to intubation attempt	
4. Recognize peri-intubation hypoxia and use BVM	